📄 Loan Recovery Suite: Operations & Propensity Engine

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Target Stakeholders: Recovery Strategy, Collections Operations, & Chief Risk Officers (CRO)

📊 Executive Summary

This interactive decisioning platform is designed to optimize the Net Present Value (NPV) of delinquent loan portfolios. By bridging the gap between raw borrower data and recovery operations, the engine provides a data-driven framework to solve the "Cure" selection dilemma: identifying exactly when to litigate an account internally versus when to execute a bulk asset sale.

Key Business Impact:

- **Yield Optimization:** Replaces "one-size-fits-all" collections with an NPV crossover model, ensuring high-propensity accounts are retained for internal recovery while low-yield assets are offloaded.
- **Risk-Adjusted Prioritization:** Leverages a Logistic Regression engine to score Propensity to Pay (PTP), allowing recovery teams to focus resources on the highest-probability targets.
- **Operational Transparency:** Features a "Human-in-the-Loop" architecture using Streamlit's `st.data_editor`, allowing managers to override system recommendations while maintaining a 100% compliant audit trail.

Access PoC

- User Name: manager
- Password: manager01$

🚀 Executive UX & Strategic Modules

The application is built for speed and precision, designed for immediate review by senior operations leads.

Module 1: Strategic Recovery Lab (NPV Crossover)

- **Goal:** Determine the financial "break-even" point for internal legal efforts.
- **Precision Modeling:** Incorporates **WACC (Cost of Capital)** and **Recovery Horizons** to discount future cash flows back to the present, comparing them directly against immediate third-party sale bids.
- **Sensitivity Analysis:** Visualizes how shifting discount rates impact the "Hold vs. Sell" decision.

Module 2: Propensity Engine (PTP & Segmentation)

- **Goal:** Segment the portfolio by behavioral and financial risk.
- **Predictive Scoring:** Uses a **Logistic Regression model** to weight Income, Days Past Due (DPD), and Balance into a singular 0–100 Propensity score.
- **Risk Matrix:** Maps Probability of Default (PD) against Loss Given Default (LGD) to categorize account health into actionable quadrants.

Module 3: Geospatial Risk & Compliance

- **Goal:** Identify regional clusters and legal friction points.
- **Mapping:** Integrates `pgeocode` for lightweight US Zip Code mapping, highlighting state-specific risks (e.g., high-difficulty litigation environments in NY or CA).

🛠️ Technical Implementation

This project demonstrates proficiency in financial engineering, predictive modeling, and interactive full-stack deployment.

- **Languages:** Python 3.x
- **Data Science:** `Pandas` (ETL), `NumPy` (NPV Modeling), `Scikit-Learn` (Logistic Regression).

- **Geospatial:** `pgeocode` (Postal mapping).
- **UI/UX:** `Streamlit` (Application framework), `Plotly Express` (Dynamic visualization).

💻 Installation & Usage

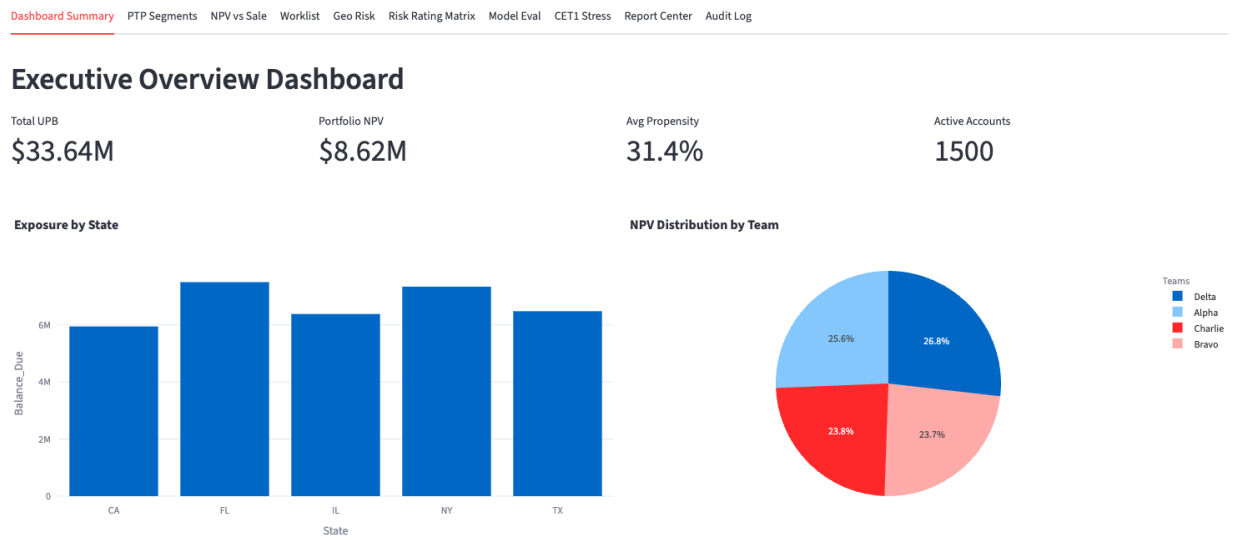
To launch the decision engine locally:

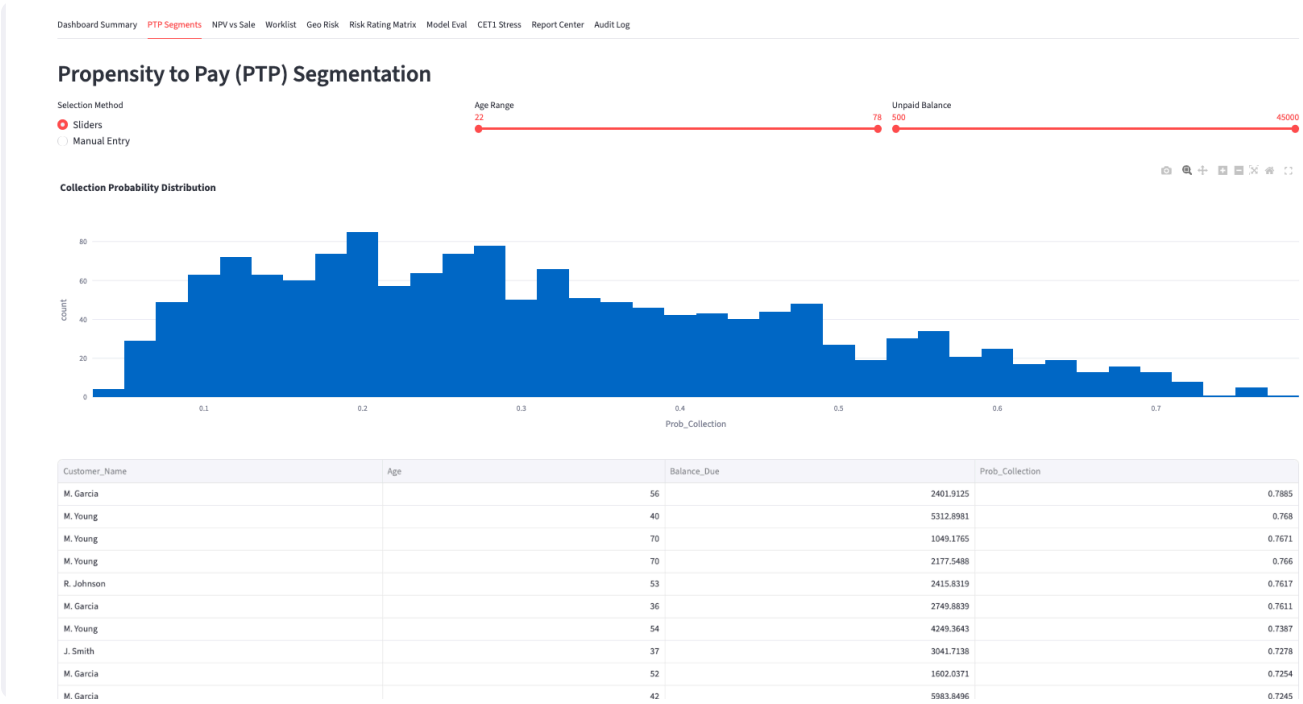
1. **Clone the repository:** `git clone https://github.com/your-username/loan-recovery-suite.git`
2. **Install dependencies:** `pip install -r requirements.txt`
3. **Run the app:** `streamlit run app.py`

📁 Repository Strategy

This repository serves as a **Narrative of Authority**, demonstrating that sophisticated 2nd-line risk strategies can only be effective when they are seamlessly integrated into 1st-line operational workflows.

Screenshots





Customer_Name	Age	Balance_Due	Prob_Collection
M. Garcia	56	2401.9125	0.7885
M. Young	40	5312.8981	0.768
M. Young	70	1049.1765	0.7671
M. Young	70	2177.5488	0.766
R. Johnson	53	2415.8319	0.7617
M. Garcia	36	2749.8839	0.7611
M. Young	54	4249.3643	0.7387
J. Smith	37	3041.7138	0.7278
M. Garcia	52	1602.0371	0.7254
M. Garcia	42	5983.8496	0.7245

Dashboard Summary PTP Segments NPV vs Sale Worklist Geo Risk Risk Rating Matrix Model Eval CET1 Stress Report Center Audit Log

Internal Recovery vs. Tranche Sale

Cost of Capital (%): 12.00

Horizon (Years): 3

Sale Price (Decimal): 0.05

Interactive Individual Account Tool

Select	Customer_Name	Balance_Due	DPD	PD_LGD_Rating	Suggested_Cure
☐	P. Miller	24852.0697	203	1D	Litigate
☐	M. Young	44633.8932	54	5B	Litigate
☐	R. Johnson	30056.9762	184	14C	Litigate
☐	M. Young	9366.445	100	2G	Litigate
☐	M. Young	2500.7842	109	3G	Asset Sale
☐	M. Garcia	6122.8634	87	14C	Litigate
☐	R. Johnson	25533.7065	173	6E	Litigate
☐	R. Johnson	34549.6346	162	3E	Litigate
☐	R. Johnson	42348.6547	137	3B	Litigate
☐	M. Young	1204.1312	8	12F	Asset Sale

Portfolio Comparison

Year	Forecasted CF	Present Value
1	$4,045,349.35	$3,611,919.06
2	$4,045,349.35	$3,224,927.73
3	$4,045,349.35	$2,879,399.76

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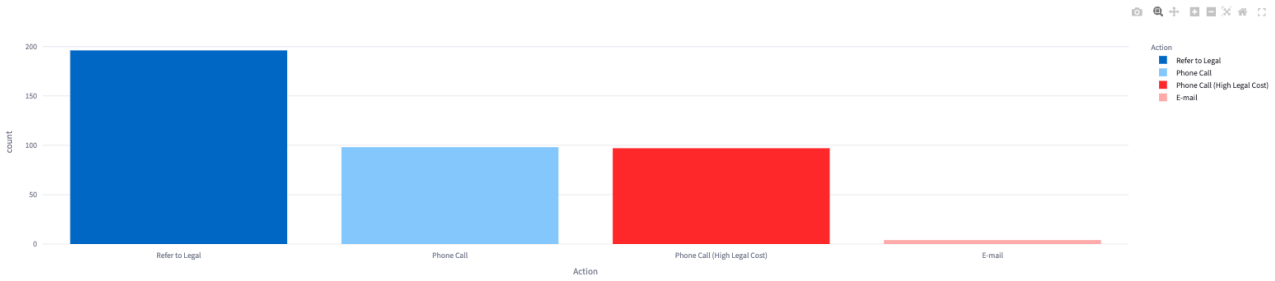
Tactical Recovery Worklist

Select Agent Team: Delta



Override	Customer_Name	Balance_Due	PD_LGD_Rating	Action	Manual_Note
☐	P. Miller	24852.0697	1D	Phone Call (High Legal Cost)	
☐	M. Young	44633.8932	5B	Refer to Legal	
☐	R. Johnson	25533.7065	6E	Phone Call (High Legal Cost)	
☐	R. Johnson	42348.6547	3B	Refer to Legal	
☐	M. Young	1204.1312	12F	Phone Call	
☐	M. Garcia	31073.2422	1C	Refer to Legal	
☐	J. Smith	23323.919	12A	Refer to Legal	
☐	P. Miller	33501.3166	1G	Phone Call (High Legal Cost)	
☐	L. King	6358.5847	8F	Phone Call	

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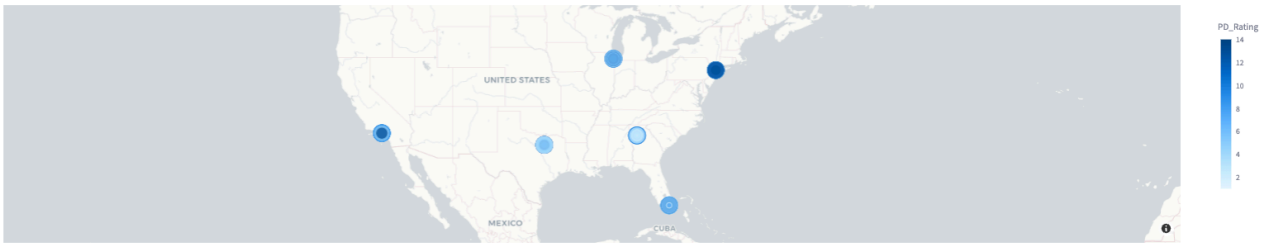
Geographic Risk Concentration



Worst PD Regions

City	State	Total Balance	Avg PD	Avg LGD
San Francisco	IL	672475.5736	8.8889	5.5185
New York	FL	665078.3007	8.8214	4.6786
Chicago	TX	672747.0311	8.8125	4.5625
Houston	NY	808208.0161	8.7778	4.1944
Dallas	FL	638888.5757	8.6471	4.1471
Los Angeles	IL	511567.3298	8.5833	4.2083
Los Angeles	CA	748126.4578	8.4828	4.1379
San Francisco	CA	458658.872	8.375	3.9167
New York	NY	684037.8543	8.3333	3.8

Best PD Regions

City	State	Total Balance	Avg PD	Avg LGD
Orlando	FL	714018.0065	5.8333	4.3333
Buffalo	FL	1324887.4379	6.0741	4.9074
Houston	FL	737952.4081	6.3333	4.9091
Naperville	CA	546109.4112	6.3571	4.3571
Naperville	IL	891695.0213	6.5	4.2333
San Francisco	FL	636784.0347	6.5385	4.3846
Chicago	NY	1013389.3334	6.7297	4.3243
Orlando	IL	816582.3978	6.8333	4.4
Los Angeles	NY	535153.1307	6.8519	4

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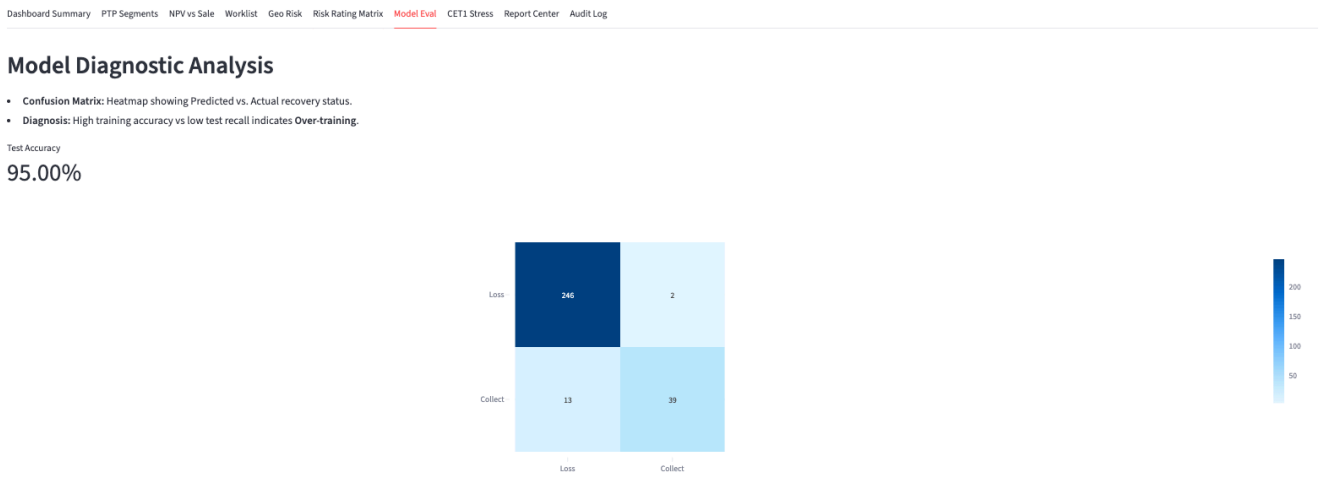
Probability of Collection Matrix

	$0-500	$500-1k	$1k-5k	$5k-10k	$10k-25k	$25k+
6 Mo	85.0%	80.8%	76.7%	72.9%	69.2%	65.8%
12 Mo	74.8%	71.1%	67.5%	64.1%	60.9%	57.9%
18 Mo	65.8%	62.5%	59.4%	56.4%	53.6%	50.9%
24 Mo	57.9%	55.0%	52.3%	49.7%	47.2%	44.8%
30 Mo	51.0%	48.4%	46.0%	43.7%	41.5%	39.4%
36 Mo	44.9%	42.6%	40.5%	38.5%	36.5%	34.7%
42 Mo	39.5%	37.5%	35.6%	33.8%	32.2%	30.5%
48 Mo	34.7%	33.0%	31.4%	29.8%	28.3%	26.9%
54 Mo	30.6%	29.0%	27.6%	26.2%	24.9%	23.7%
60 Mo	26.9%	25.6%	24.3%	23.1%	21.9%	20.8%

Account Details

Account_ID	Balance_Due	PD_Rating	LGD_Grade	NPV_Owed	Suggested_Cure
ACC-10000	24852.0697	1	D	2781.3408	Litigate
ACC-10001	44633.8932	5	B	11160.6966	Litigate
ACC-10002	30056.9762	14	C	4780.7375	Litigate
ACC-10003	9366.445	2	G	4084.7953	Litigate
ACC-10004	2500.7842	3	G	1397.961	Asset Sale
ACC-10005	6122.8634	14	C	3331.2408	Litigate
ACC-10006	25533.7065	6	E	4057.0452	Litigate
ACC-10007	34549.6346	3	E	5039.209	Litigate
ACC-10008	42348.6547	3	B	4014.9642	Litigate
ACC-10009	1204.1312	12	F	841.6425	Asset Sale


Dashboard Summary
PTP Segments
NPV vs Sale
Worklist
Geo Risk
Risk Rating Matrix
Model Eval
CET1 Stress
Report Center
Audit Log
Model Diagnostic Analysis
Confusion Matrix: Heatmap showing Predicted vs. Actual recovery status.
Diagnosis: High training accuracy vs low test recall indicates Over-training.
Test Accuracy
95.00%
Loss
Collect
246
2
13
39
Loss
Collect
200
150
100
50

Dashboard Summary
PTP Segments
NPV vs Sale
Worklist
Geo Risk
Risk Rating Matrix
Model Eval
CET1 Stress
Report Center
Audit Log
CET1: RWA Reduction Analysis
Input Mode
Slider
Manual Entry
Target %
13.00
RWA Reduction Required
$3,234,198
RWA Sell-off vs. Target %
RWA Reduction Needed ($)
10M
5M
0
8
10
12
14
16
Target Ratio (%)

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Executive Reports

Report

ROI Analysis

Agent_Team	Balance_Due	NPV_Owed	Prob_Collection	Efficiency_Score
Alpha	$8,756,059.06	$2,210,156.26	31.2%	25.24%
Bravo	$7,855,022.69	$2,047,715.70	32.4%	26.07%
Charlie	$8,272,116.57	$2,053,944.09	30.1%	24.83%
Delta	$8,752,464.00	$2,311,160.86	31.9%	26.41%

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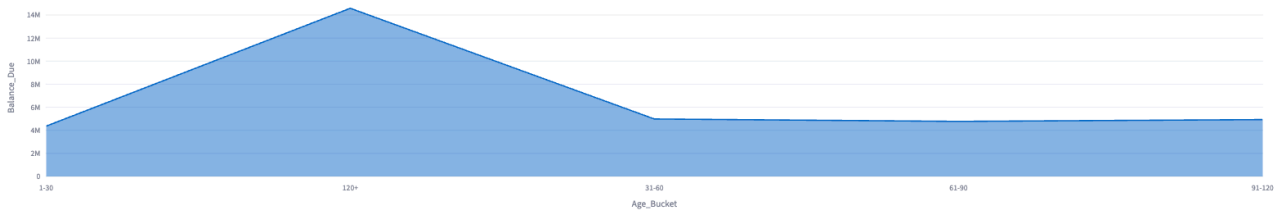
Executive Reports

Report

Vintage Decay

Exposure Decay by Aging Bucket



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Executive Reports

Report

Compliance

Account_ID	State	Original_Disposition	PD_LGD_Rating
ACC-10000	CA	Legal Remedy	1D
ACC-10001	FL	Legal Remedy	5B
ACC-10002	FL	Legal Remedy	14C
ACC-10003	CA	Legal Remedy	2G
ACC-10004	CA	Asset Sale	3G
ACC-10005	TX	Legal Remedy	14C
ACC-10006	CA	Legal Remedy	6E
ACC-10007	TX	Legal Remedy	3E
ACC-10008	TX	Legal Remedy	3B
ACC-10009	TX	Asset Sale	12F

Dashboard Summary PTP Segments NPV vs Sale Worklist Geo Risk Risk Rating Matrix Model Eval CET1 Stress Report Center Audit Log

Audit Log

Timestamp	Borrower	Balance	PD_LGD	Recommended_Action	Manual_Override	Note
2026-04-08 18:43:50	M. Young	44633.8932	5B	Refer to Legal	Manager Decision	
2026-04-08 18:43:50	R. Johnson	42348.6547	3B	Refer to Legal	Manager Decision	
2026-04-08 18:43:50	M. Garcia	31073.2422	1C	Refer to Legal	Manager Decision	
2026-04-08 18:43:50	J. Smith	23323.919	12A	Refer to Legal	Manager Decision	
2026-04-08 18:43:50	L. King	38534.3035	4F	Refer to Legal	Manager Decision	
2026-04-08 18:43:50	P. Miller	22339.0428	3A	Refer to Legal	Manager Decision	